Incompressible Stokes Flow Annulus Benchmark Using Underworld3

Thyagarajulu Gollapalli[1] and Underworld Team[1]

[1]Research School of Earth Sciences, Australian National University, Canberra, ACT, Australia

Correspondence: Thyagarajulu Gollapalli (gthyagi@gmail.com)

Abstract. We reproduce the annulus Stokes benchmark of Thieulot and Puckett (2018) (TP2018) using Underworld3 (UW3). The benchmark describes incompressible isoviscous Stokes flow in a cylindrical annulus with tangential velocity boundary conditions imposed on both the inner and outer boundaries, driven by radially directed gravity and analytically prescribed buoyancy forcing that generates convection cells parameterised by the harmonic number k. We investigate the accuracy and convergence behaviour of several mixed finite-element discretisations, including $P_1 \times P_0$, $P_1 \times P_1$, $P_2 \times P_0$, $P_2 \times P_{-1}$, $P_2 \times P_1$, and $P_3 \times P_2$. Numerical solutions are evaluated against the analytical velocity and pressure fields using relative L_2-norm errors. The higher-order Taylor–Hood discretisations reproduce the expected convergence hierarchy for smooth Stokes solutions, with approximately third-order velocity and second-order pressure convergence for $P_2 \times P_1$, and fourth-order velocity and third-order pressure convergence for $P_3 \times P_2$. In contrast, lower-order discretisations exhibit reduced robustness, particularly in pressure recovery. These results demonstrate that UW3 accurately reproduces analytical Stokes solutions in curved geometries and provides a robust framework for future annulus geodynamics simulations.

1 Introduction

Accurate numerical solution of the incompressible Stokes equations is fundamental to modelling mantle convection, lithospheric deformation, subduction dynamics, and planetary evolution. Many geodynamic applications require simulations in curved geometries, such as annulus, spherical cap, and full spherical shell. These geometries introduce additional numerical challenges associated with curved boundaries, pressure nullspaces, coordinate-dependent forcing terms, and tangential boundary conditions.

Reproducing analytical benchmarks remains one of the most rigorous approaches for assessing the correctness of numerical implementations. Unlike code-to-code comparisons, analytical benchmarks enable direct quantification of discretisation errors and convergence rates. They are particularly valuable for geodynamic Stokes problems because they simultaneously assess velocity–pressure coupling, incompressibility preservation, boundary condition implementation, and the accurate representation of curved geometries (Kramer et al., 2021; Davies et al., 2022).

The annulus benchmark introduced by TP2018 provides an exact analytical solution for incompressible isoviscous Stokes flow in a cylindrical shell subjected to radially directed gravity. The benchmark was specifically designed for geodynamics applications and has since been widely used for verification of geodynamic codes within the community (Gassmöller et al., 2024).

Unlike classical Cartesian manufactured solutions, the benchmark incorporates curved inner and outer boundaries, non-trivial pressure distributions along the boundaries, coordinate-dependent buoyancy forcing, tangential velocity boundary conditions, and controllable azimuthal complexity through the harmonic parameter k. These characteristics make the benchmark particularly well suited for assessing finite-element formulations in annulus and curved-domain mantle convection problems.

In this study, we reproduce the TP2018 annulus benchmark to verify the curved-domain finite-element implementations in UW3. We evaluate the convergence and accuracy of several mixed finite-element discretisations, including both lower-order and higher-order velocity–pressure element pairs. Numerical results are compared against the analytical velocity and pressure fields using L_2-norm errors and boundary pressure diagnostics. Particular attention is given to the convergence behaviour of Taylor–Hood discretisations and the performance of lower-order and equal-order element formulations.

2 Governing Equations

We consider steady state incompressible Stokes equations:

$$-\nabla \cdot \boldsymbol{\sigma} = \rho \mathbf{g}, \tag{1a}$$

$$\nabla \cdot \mathbf{u} = 0. \tag{1b}$$

The stress tensor is

$$\boldsymbol{\sigma} = -p\mathbf{I} + 2\eta\boldsymbol{\varepsilon}(\mathbf{u}), \tag{2}$$

with viscosity η and strain-rate tensor

$$\boldsymbol{\varepsilon}(\mathbf{u}) = \frac{1}{2}\left(\nabla\mathbf{u} + (\nabla\mathbf{u})^T\right). \tag{3}$$

where $\mathbf{u}$ is velocity, p is pressure, ρ is density, and $\mathbf{g}$ is gravitational acceleration.

3 Analytical Benchmark Solution

Following Thieulot and Puckett (2018), the analytical velocity field in polar coordinates is defined as

$$v_\theta(r,\theta) = f(r)\cos(k\theta), \tag{4a}$$

$$v_r(r,\theta) = kg(r)\sin(k\theta). \tag{4b}$$

where

$$f(r) = Ar + \frac{B}{r}, \tag{5a}$$

$$g(r) = \frac{A}{2}r + \frac{B}{r}\ln r + \frac{C}{r}. \tag{5b}$$

The constants A and B are chosen such that radial velocity vanishes at both annulus boundaries:

$$A = -C\frac{2(\ln R_1 - \ln R_2)}{R_2^2 \ln R_1 - R_1^2 \ln R_2}, \tag{6a}$$

$$B = -C\frac{R_2^2 - R_1^2}{R_2^2 \ln R_1 - R_1^2 \ln R_2}. \tag{6b}$$

The pressure field is

$$p(r,\theta) = kh(r)\sin(k\theta) + \rho_0(R_2 - r), \tag{7}$$

where

$$h(r) = \frac{2g(r) - f(r)}{r}. \tag{8}$$

The corresponding density field is

$$\rho(r,\theta) = kM(r)\sin(k\theta) + \rho_0, \tag{9}$$

with

$$M(r) = g''(r) - \frac{g'(r)}{r} - \frac{g(r)}{r^2}(k^2 - 1) + \frac{f(r)}{r^2} + \frac{f'(r)}{r}. \tag{10}$$

The benchmark calculations use the standard parameter values

$$R_1 = 1, \qquad R_2 = 2, \qquad C = -1, \qquad \rho_0 = 0.$$

where R_1 is the inner radius and R_2 is the outer radius. Gravity acts radially inward:

$$\mathbf{g} = -\mathbf{e}_r. \tag{11}$$

The benchmark assumes constant viscosity $\eta = 1$ and prescribes analytical velocity on boundaries, and density field in the domain. Increasing the harmonic parameter k increases azimuthal complexity and produces $2k$ convection cells in the annulus. This enables systematic evaluation of numerical performance across increasingly oscillatory flow structures.

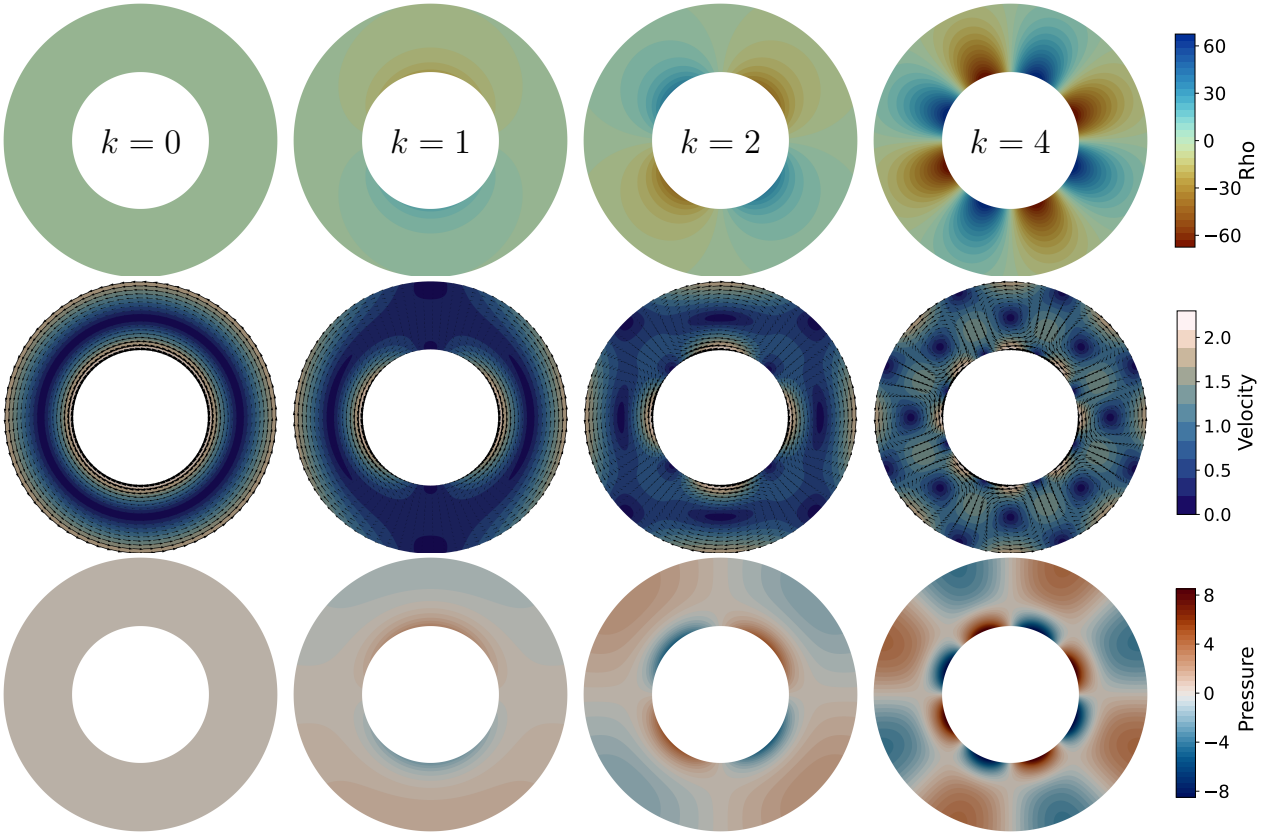

Figure 1. Analytical density, velocity, and pressure fields. Columns correspond to increasing k, which increases the number of convection cells in the annulus.

4 Numerical Methodology

4.1 UW3 Formulation

The benchmark is implemented in Underworld3 (UW3), a next-generation geodynamics framework built on PETSc and symbolic problem specification in Python. The incompressible Stokes system is discretised using the finite-element method and assembled through symbolic expressions defined with SymPy. The resulting linear systems are solved using PETSc Krylov subspace solvers and preconditioners.

To remove the pressure gauge ambiguity inherent to incompressible Stokes flow, a constant-pressure nullspace is supplied to PETSc during the solve. Following solution, the domain-average pressure is removed from the numerical solution prior to error evaluation and comparison with the analytical benchmark solution.

The annulus domain is discretised using unstructured triangular meshes generated with progressively refined nominal cell sizes. All simulations reported in this study employ linear geometric representation of the curved annulus boundaries.

4.2 Finite-Element Discretisations

The following mixed finite-element pairs are evaluated:

Velocity space	Pressure space	Description
P_1	P_0	Linear velocity, discontinuous constant pressure
P_1	P_1	Equal-order continuous discretisation
P_2	P_0	Quadratic velocity, discontinuous constant pressure
P_2	P_{-1}	Quadratic velocity, discontinuous linear pressure
P_2	P_1	Taylor–Hood discretisation
P_3	P_2	Higher-order Taylor–Hood discretisation

The quadrature degree is chosen according to the highest polynomial order present in the discretisation. For smooth incompressible Stokes solutions, stable Taylor–Hood-type discretisations are expected to exhibit velocity convergence of order $p+1$ and pressure convergence of order p for a $P_p \times P_{p-1}$ discretisation (Girault and Raviart, 1986; Boffi et al., 2013).

4.3 Error Metrics

Errors are quantified by comparing the numerical and analytical solutions using both absolute and relative L_2-norm measures. The absolute L_2-norm error is defined as

$$E_{L_2}(q) = \left(\int_\Omega |q - q^*|^2 \, d\Omega \right)^{1/2}, \tag{12}$$

and the relative L_2-norm error is computed as

$$E_{L_2}^{\mathrm{rel}}(q) = \frac{E_{L_2}(q)}{\|q^*\|_{L_2}} = \left(\frac{\int_\Omega |q - q^*|^2 \, d\Omega}{\int_\Omega |q^*|^2 \, d\Omega} \right)^{1/2}, \tag{13}$$

where q and q^* denote the numerical and analytical solutions, respectively.

Convergence rates are evaluated from successive mesh refinements as

$$rate = \frac{\log\left(E_{L_2,h_1}(q)/E_{L_2,h_2}(q)\right)}{\log\left(h_1/h_2\right)}, \tag{14}$$

where h_1 and h_2 are the characteristic mesh sizes of two consecutive refinements. For uniform mesh refinement with $h_2 = h_1/2$, this reduces to

$$rate = \log_2 \left(\frac{E_{L_2,h_1}(q)}{E_{L_2,h_2}(q)} \right). \tag{15}$$

5 Results

5.1 Convergence Behaviour

The convergence behaviour for all tested discretisations is shown in Fig. 2. The $P_2 \times P_1$ Taylor–Hood discretisation exhibits approximately third-order velocity convergence and second-order pressure convergence, consistent with theoretical expectations for smooth Stokes solutions.

The higher-order $P_3 \times P_2$ discretisation achieves approximately fourth-order velocity convergence and third-order pressure convergence on coarse-to-intermediate mesh resolutions. At the finest resolutions, convergence rates begin to plateau, likely reflecting accumulated effects from geometry approximation, quadrature errors, and solver tolerances.

The lower-order discretisations show reduced robustness. The $P_1 \times P_0$ pair fails to produce reliable convergence behaviour for this benchmark, while the equal-order $P_1 \times P_1$ discretisation produces acceptable velocity convergence but weaker and less consistent pressure convergence. The discontinuous-pressure $P_2 \times P_0$ pair gives approximately second-order velocity convergence and first-order pressure convergence, whereas $P_2 \times P_{-1}$ gives near second-order velocity convergence but poor pressure convergence over the available mesh refinements. These observations are consistent with established finite-element theory for incompressible Stokes flow (Boffi et al., 2013; Thieulot and Bangerth, 2025).

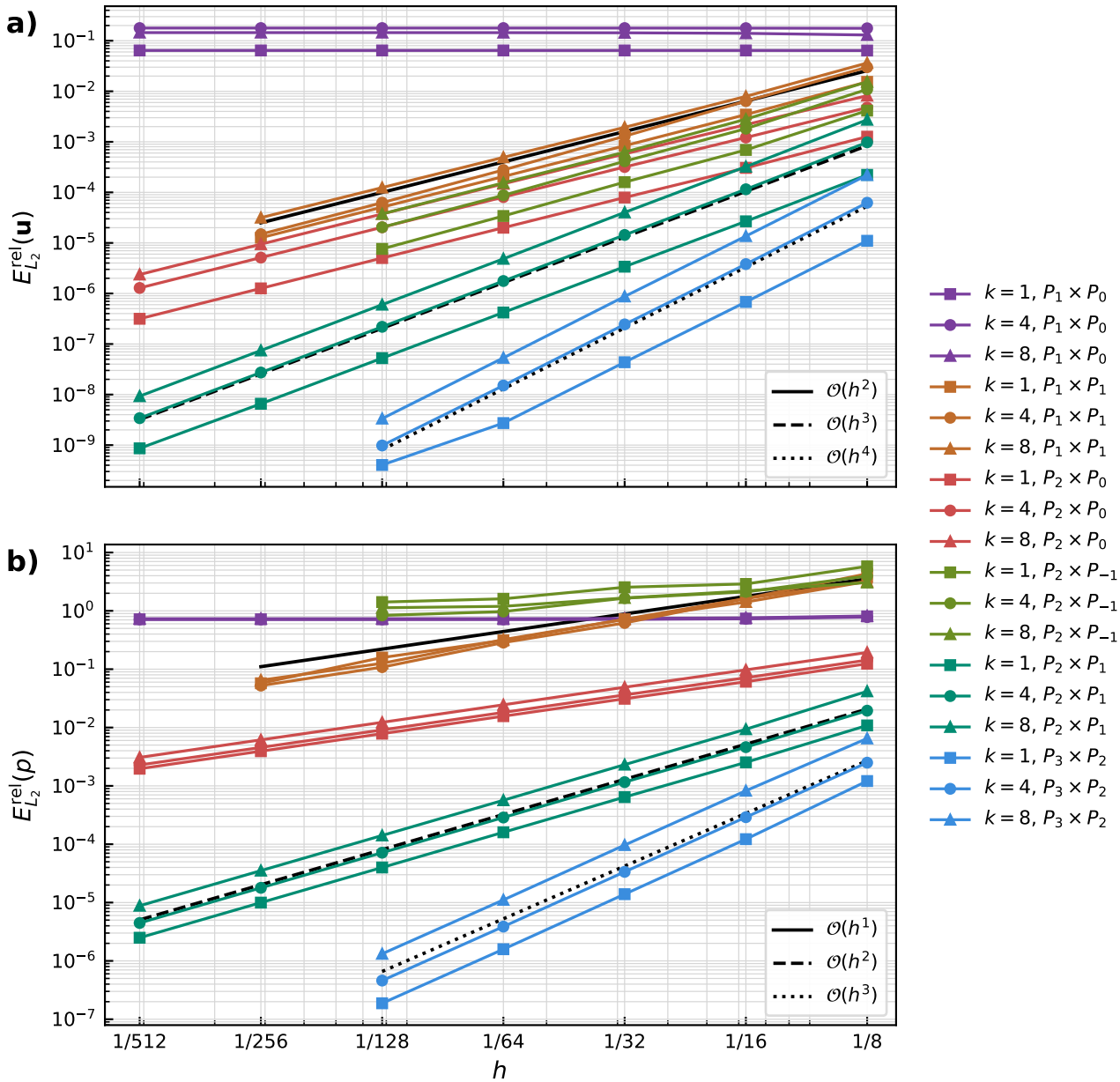

Figure 2. Convergence of relative volume errors $E^{\mathrm{rel}}_{L_2}(\mathbf{u})$ and $E^{\mathrm{rel}}_{L_2}(p)$ for the annulus Thieulot benchmark. Reference lines indicate the expected smooth-solution rates for the higher-order element pairs.

Table 1. Relative errors and convergence rates for $P_1 \times P_0$.

	$k=1$				$k=4$				$k=8$			
h	$E_{L_2}^{\mathrm{rel}}(\mathbf{u})$	rate	$E_{L_2}^{\mathrm{rel}}(p)$	rate	$E_{L_2}^{\mathrm{rel}}(\mathbf{u})$	rate	$E_{L_2}^{\mathrm{rel}}(p)$	rate	$E_{L_2}^{\mathrm{rel}}(\mathbf{u})$	rate	$E_{L_2}^{\mathrm{rel}}(p)$	rate
$1/8$	6.398e-02	–	8.040e-01	–	1.763e-01	–	7.800e-01	–	1.298e-01	–	8.163e-01	–
$1/16$	6.427e-02	-0.01	7.384e-01	0.12	1.775e-01	-0.01	7.295e-01	0.10	1.401e-01	-0.11	7.589e-01	0.11
$1/32$	6.426e-02	0.00	7.220e-01	0.03	1.780e-01	0.00	7.161e-01	0.03	1.435e-01	-0.04	7.403e-01	0.04
$1/64$	6.426e-02	0.00	7.177e-01	0.01	1.781e-01	0.00	7.124e-01	0.01	1.445e-01	-0.01	7.349e-01	0.01
$1/128$	6.425e-02	0.00	7.167e-01	0.00	1.782e-01	0.00	7.114e-01	0.00	1.447e-01	0.00	7.334e-01	0.00
$1/256$	6.425e-02	0.00	7.164e-01	0.00	1.782e-01	0.00	7.112e-01	0.00	1.448e-01	0.00	7.330e-01	0.00
$1/512$	6.425e-02	0.00	7.163e-01	0.00	1.782e-01	0.00	7.111e-01	0.00	1.448e-01	0.00	7.329e-01	0.00

Table 2. Relative errors and convergence rates for $P_1 \times P_1$.

	$k=1$				$k=4$				$k=8$			
h	$E_{L_2}^{\mathrm{rel}}(\mathbf{u})$	rate	$E_{L_2}^{\mathrm{rel}}(p)$	rate	$E_{L_2}^{\mathrm{rel}}(\mathbf{u})$	rate	$E_{L_2}^{\mathrm{rel}}(p)$	rate	$E_{L_2}^{\mathrm{rel}}(\mathbf{u})$	rate	$E_{L_2}^{\mathrm{rel}}(p)$	rate
$1/8$	1.537e-02	–	4.325e+00	–	2.994e-02	–	3.202e+00	–	3.629e-02	–	3.184e+00	–
$1/16$	3.459e-03	2.15	1.628e+00	1.41	6.385e-03	2.23	1.625e+00	0.98	7.925e-03	2.19	1.417e+00	1.17
$1/32$	8.358e-04	2.05	7.174e-01	1.18	1.293e-03	2.30	6.195e-01	1.39	1.937e-03	2.03	7.144e-01	0.99
$1/64$	2.056e-04	2.02	3.115e-01	1.20	2.776e-04	2.22	2.855e-01	1.12	4.931e-04	1.97	3.210e-01	1.15
$1/128$	5.095e-05	2.01	1.582e-01	0.98	6.252e-05	2.15	1.083e-01	1.40	1.243e-04	1.99	1.260e-01	1.35
$1/256$	1.266e-05	2.01	5.740e-02	1.46	1.487e-05	2.07	5.231e-02	1.05	3.158e-05	1.98	6.476e-02	0.96

Table 3. Relative errors and convergence rates for $P_2 \times P_0$.

	$k=1$				$k=4$				$k=8$			
h	$E_{L_2}^{\mathrm{rel}}(\mathbf{u})$	rate	$E_{L_2}^{\mathrm{rel}}(p)$	rate	$E_{L_2}^{\mathrm{rel}}(\mathbf{u})$	rate	$E_{L_2}^{\mathrm{rel}}(p)$	rate	$E_{L_2}^{\mathrm{rel}}(\mathbf{u})$	rate	$E_{L_2}^{\mathrm{rel}}(p)$	rate
$1/8$	1.266e-03	–	1.239e-01	–	4.755e-03	–	1.431e-01	–	8.194e-03	–	1.934e-01	–
$1/16$	3.050e-04	2.05	6.095e-02	1.02	1.216e-03	1.97	7.140e-02	1.00	2.187e-03	1.91	9.733e-02	0.99
$1/32$	7.908e-05	1.95	3.110e-02	0.97	3.177e-04	1.94	3.619e-02	0.98	5.761e-04	1.92	4.902e-02	0.99
$1/64$	1.998e-05	1.98	1.564e-02	0.99	8.066e-05	1.98	1.814e-02	1.00	1.474e-04	1.97	2.453e-02	1.00
$1/128$	5.023e-06	1.99	7.846e-03	1.00	2.039e-05	1.98	9.099e-03	1.00	3.737e-05	1.98	1.228e-02	1.00
$1/256$	1.260e-06	1.99	3.931e-03	1.00	5.127e-06	1.99	4.557e-03	1.00	9.428e-06	1.99	6.147e-03	1.00
$1/512$	3.157e-07	2.00	1.967e-03	1.00	1.286e-06	2.00	2.280e-03	1.00	2.368e-06	1.99	3.076e-03	1.00

Table 4. Relative errors and convergence rates for $P_2 \times P_{-1}$.

	$k=1$				$k=4$				$k=8$			
h	$E_{L_2}^{\mathrm{rel}}(\mathbf{u})$	rate	$E_{L_2}^{\mathrm{rel}}(p)$	rate	$E_{L_2}^{\mathrm{rel}}(\mathbf{u})$	rate	$E_{L_2}^{\mathrm{rel}}(p)$	rate	$E_{L_2}^{\mathrm{rel}}(\mathbf{u})$	rate	$E_{L_2}^{\mathrm{rel}}(p)$	rate
$1/8$	4.143e-03	–	5.788e+00	–	1.093e-02	–	3.954e+00	–	1.538e-02	–	3.054e+00	–
$1/16$	6.912e-04	2.58	2.890e+00	1.00	1.821e-03	2.59	2.111e+00	0.91	2.802e-03	2.46	2.193e+00	0.48
$1/32$	1.594e-04	2.12	2.538e+00	0.19	4.121e-04	2.14	1.648e+00	0.36	6.249e-04	2.16	1.663e+00	0.40
$1/64$	3.410e-05	2.22	1.608e+00	0.66	8.750e-05	2.24	9.717e-01	0.76	1.543e-04	2.02	1.195e+00	0.48
$1/128$	7.613e-06	2.16	1.414e+00	0.19	2.043e-05	2.10	8.349e-01	0.22	3.752e-05	2.04	1.129e+00	0.08

Table 5. Relative errors and convergence rates for $P_2 \times P_1$.

	$k=1$				$k=4$				$k=8$			
h	$E_{L_2}^{\mathrm{rel}}(\mathbf{u})$	rate	$E_{L_2}^{\mathrm{rel}}(p)$	rate	$E_{L_2}^{\mathrm{rel}}(\mathbf{u})$	rate	$E_{L_2}^{\mathrm{rel}}(p)$	rate	$E_{L_2}^{\mathrm{rel}}(\mathbf{u})$	rate	$E_{L_2}^{\mathrm{rel}}(p)$	rate
$1/8$	2.218e-04	–	1.087e-02	–	9.781e-04	–	1.944e-02	–	2.731e-03	–	4.174e-02	–
$1/16$	2.675e-05	3.05	2.530e-03	2.10	1.151e-04	3.09	4.611e-03	2.08	3.225e-04	3.08	9.354e-03	2.16
$1/32$	3.388e-06	2.98	6.427e-04	1.98	1.437e-05	3.00	1.159e-03	1.99	4.016e-05	3.01	2.310e-03	2.02
$1/64$	4.205e-07	3.01	1.601e-04	2.01	1.763e-06	3.03	2.869e-04	2.01	4.850e-06	3.05	5.666e-04	2.03
$1/128$	5.254e-08	3.00	3.992e-05	2.00	2.190e-07	3.01	7.152e-05	2.00	6.001e-07	3.01	1.412e-04	2.00
$1/256$	6.575e-09	3.00	9.979e-06	2.00	2.730e-08	3.00	1.787e-05	2.00	7.449e-08	3.01	3.524e-05	2.00
$1/512$	8.632e-10	2.93	2.494e-06	2.00	3.412e-09	3.00	4.466e-06	2.00	9.287e-09	3.00	8.802e-06	2.00

Table 6. Relative errors and convergence rates for $P_3 \times P_2$.

	$k=1$				$k=4$				$k=8$			
h	$E_{L_2}^{\mathrm{rel}}(\mathbf{u})$	rate	$E_{L_2}^{\mathrm{rel}}(p)$	rate	$E_{L_2}^{\mathrm{rel}}(\mathbf{u})$	rate	$E_{L_2}^{\mathrm{rel}}(p)$	rate	$E_{L_2}^{\mathrm{rel}}(\mathbf{u})$	rate	$E_{L_2}^{\mathrm{rel}}(p)$	rate
$1/8$	1.102e-05	–	1.210e-03	–	6.233e-05	–	2.498e-03	–	2.198e-04	–	6.495e-03	–
$1/16$	6.792e-07	4.02	1.220e-04	3.31	3.811e-06	4.03	2.905e-04	3.10	1.363e-05	4.01	8.249e-04	2.98
$1/32$	4.360e-08	3.96	1.392e-05	3.13	2.441e-07	3.96	3.358e-05	3.11	8.723e-07	3.97	9.664e-05	3.09
$1/64$	2.729e-09	4.00	1.585e-06	3.13	1.510e-08	4.01	3.870e-06	3.12	5.323e-08	4.03	1.110e-05	3.12
$1/128$	4.043e-10	2.75	1.877e-07	3.08	9.826e-10	3.94	4.601e-07	3.07	3.344e-09	3.99	1.322e-06	3.07

5.2 Boundary Pressure Recovery

Pressure profiles extracted along the inner and outer annulus boundaries are compared against analytical solutions in Fig. 3. The higher-order Taylor–Hood discretisations reproduce the analytical pressure distributions accurately across the annulus boundaries, including extrema near regions of high azimuthal variation. Pressure discrepancies are most pronounced for lower-order discretisations and become increasingly visible for larger k, where sharper angular pressure gradients must be resolved.

Because the benchmark includes non-trivial pressure variation directly on curved boundaries, these comparisons provide a stringent test of both pressure interpolation quality and curved-domain numerical accuracy.

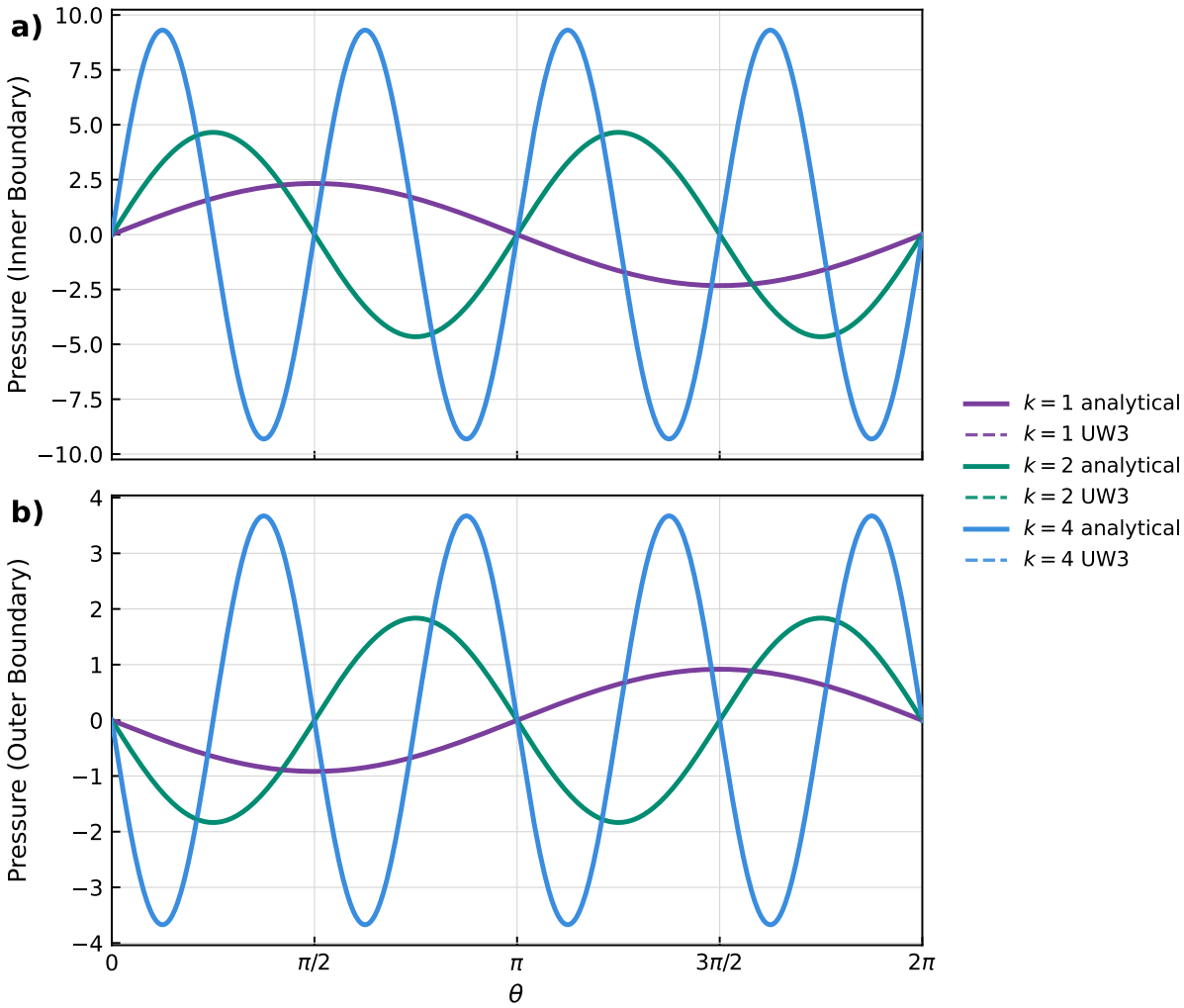

Figure 3. Pressure along the inner and outer annulus boundaries for $k = 1, 2, 4$, using the $P_2 \times P_1$, $h = 1/128$ calculations. Solid curves denote the analytical solution and dashed curves denote UW3 values sampled from the numerical pressure field.

The corresponding absolute boundary pressure errors for the $P_2 \times P_1$ discretisation are shown in Fig. 4. Both inner- and outer-boundary errors decrease at approximately second order, consistent with the volume pressure convergence observed for this element pair.

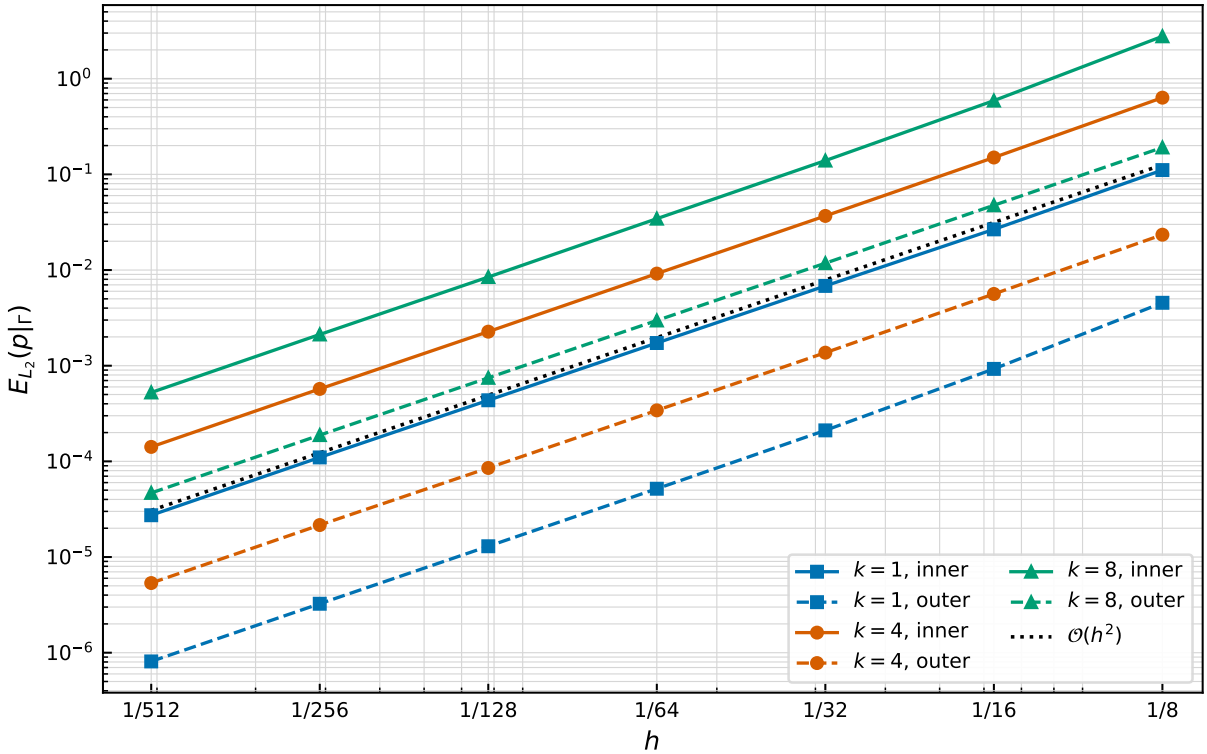

Figure 4. Convergence of absolute boundary pressure errors $E_{L_2}(p|_\Gamma)$ for the $P_2 \times P_1$ annulus Thieulot benchmark calculations. Solid curves denote the inner boundary and dashed curves denote the outer boundary.

6 Discussion

The present results demonstrate that Underworld3 reproduces the analytical annulus benchmark solution with convergence behaviour consistent with finite-element theory. The benchmark additionally highlights several important numerical considerations.

Table 7 summarises the observed convergence rates for the three harmonic benchmark cases for which analytical solutions are available. The optimal-rate column reports the smooth-solution reference rates used here; entries marked $-/-$ have no optimal-rate claim in this comparison, and their fitted rates should be interpreted empirically.

Table 7. Observed convergence rates for the three annulus benchmark cases with analytical solutions. The optimal column gives the expected smooth-solution rates used here as velocity/pressure; $-/-$ denotes element pairs for which no optimal-rate claim is made here. Observed entries are least-squares slopes fitted to $\log E$ versus $\log h$ and are reported as $E^{\mathrm{rel}}_{L_2}(\mathbf{u})/E^{\mathrm{rel}}_{L_2}(p)$.

Element pair	Status	Optimal	$k=1$	$k=4$	$k=8$
$P_1 \times P_0$	unstable	-/-	0.0/0.0	0.0/0.0	0.0/0.0
$P_1 \times P_1$	non-standard	-/-	2.0/1.2	2.2/1.2	2.0/1.1
$P_2 \times P_0$	non-standard	2/1	2.0/1.0	2.0/1.0	2.0/1.0
$P_2 \times P_{-1}$	non-standard	-/-	2.3/0.5	2.3/0.6	2.2/0.4
$P_2 \times P_1$	stable	3/2	3.0/2.0	3.0/2.0	3.0/2.0
$P_3 \times P_2$	stable	4/3	3.7/3.2	4.0/3.1	4.0/3.1

The stability status of the lower-order and discontinuous-pressure pairs requires some qualification. The equal-order $P_1 \times P_1$ pair is not inf-sup stable as a standard Galerkin Stokes discretisation and is therefore susceptible to spurious pressure modes unless an additional pressure-stabilisation mechanism is introduced (Boffi et al., 2013). Stabilised equal-order formulations, such as Petrov–Galerkin, Galerkin least-squares, or related stabilised mixed methods, can restore well-posedness and recover the smooth-solution target rates $E_{L_2}(\mathbf{u})/E_{L_2}(p) = 2/1$ (Hughes et al., 1986; Brezzi and Douglas, 1988). The $P_2 \times P_0$ pair is inf-sup stable, but is non-standard and approximation-unbalanced: the constant pressure space limits the expected pressure convergence to first order and the velocity convergence to second order (Boffi et al., 2013; Thieulot and Bangerth, 2025). The $P_2 \times P_{-1}$ pair used here denotes continuous quadratic velocity with discontinuous linear pressure, i.e. the lowest-order Scott–Vogelius-type $P_2 \times P_1^{\mathrm{disc}}$ pairing. This pair is not uniformly stable on arbitrary triangular meshes, but can be made stable on suitable split meshes, including barycentric/Alfeld refinements, where the smooth-solution target is $3/2$ (Scott and Vogelius, 1985; Guzmán and Neilan, 2018).

Table 8. Stability interpretation and expected smooth-solution rates for selected non-Taylor–Hood element pairs. Rates are reported as $E_{L_2}(\mathbf{u})/E_{L_2}(p)$.

Pair	Status in standard Galerkin form	Route to stable use	Expected rate
$P_1 \times P_1$	Unstable equal-order pair	Add consistent pressure stabilisation, e.g. PSPG/GLS or related stabilised mixed methods	$2/1$ when stabilised
$P_2 \times P_0$	Stable but non-standard and pressure-limited	No stabilisation required, but pressure approximation remains piecewise constant	$2/1$
$P_2 \times P_{-1}$	Not uniformly stable on arbitrary triangular meshes	Use suitable Scott–Vogelius meshes such as barycentric/Alfeld refinements, or equivalent stabilising enrichment/restriction	$3/2$ when stable

6.1 Boundary Pressure Error Asymmetry

The inner- and outer-boundary pressure errors in Fig. 4 are not expected to be equal, even though the analytical pressure has zero signed integral on each boundary. With $\rho_0 = 0$, the analytical boundary pressure is

$$p^*(r,\theta) = kh(r)\sin(k\theta), \tag{16}$$

and therefore

$$\int_{\Gamma_r} p^*\, ds = \int_0^{2\pi} kh(r)\sin(k\theta)\, r\, d\theta = 0. \tag{17}$$

The error plotted in Fig. 4 is instead an absolute boundary L_2 error,

$$E_{L_2}(p|_{\Gamma_r}) = \left(\int_{\Gamma_r} |p_h - p^*|^2\, ds \right)^{1/2} = \left(\int_0^{2\pi} |p_h(r,\theta) - p^*(r,\theta)|^2 r\, d\theta \right)^{1/2}. \tag{18}$$

The corresponding analytical pressure magnitude is

$$\|p^*\|_{L_2(\Gamma_r)} = |kh(r)|\sqrt{\pi r}, \tag{19}$$

which is different on the two boundaries. For the present benchmark parameters,

$$h(R_1) \simeq 2.328, \qquad h(R_2) \simeq -0.918, \qquad \frac{\|p^*\|_{L_2(\Gamma_{R_2})}}{\|p^*\|_{L_2(\Gamma_{R_1})}} \simeq 0.558.$$

Thus the signed mean cancels on both boundaries, but the pressure trace has larger L_2 magnitude on the inner boundary.

A second source of asymmetry is the sensitivity of the pressure trace to small geometric perturbations. If a discrete boundary point is displaced from the exact circular boundary by

$$x_h(\theta) - x(\theta) = \delta r(\theta)\mathbf{e}_r + r\,\delta\theta(\theta)\mathbf{e}_\theta,$$

then the analytical pressure sampled on the perturbed boundary satisfies, to first order,

$$p^*(r+\delta r, \theta+\delta\theta) - p^*(r,\theta) \simeq kh'(r)\sin(k\theta)\,\delta r + k^2 h(r)\cos(k\theta)\,\delta\theta. \tag{20}$$

Consequently, a boundary-pressure error model contains the geometric component

$$E_{L_2}^2(p|_{\Gamma_r}) \approx r \int_0^{2\pi} \left| e_{\mathrm{disc}}(\theta) + kh'(r)\sin(k\theta)\,\delta r + k^2 h(r)\cos(k\theta)\,\delta\theta \right|^2 d\theta, \tag{21}$$

where e_{disc} contains the pressure discretisation and solver errors on the represented boundary. For the benchmark constants,

$$h'(R_1) \simeq -13.31, \qquad h'(R_2) \simeq -0.164.$$

The inner pressure trace is therefore far more sensitive to small radial boundary-location errors than the outer trace. This explains why the absolute inner-boundary error can be much larger than the outer-boundary error, despite the outer boundary having a larger arclength measure.

Linear boundary elements approximate a circular boundary by straight chords. If Δ denotes the boundary mesh spacing, the geometric displacement is typically $O(\Delta^2)$, while the normal error is $O(\Delta)$. Classical curved-element and isoparametric finite-element estimates show how higher-order geometry can recover the optimal approximation behaviour expected on smooth domains (Ciarlet and Raviart, 1972; Lenoir, 1986; Bernardi, 1989). A curved isoparametric element of geometric degree m uses

$$x_h(\xi) = \sum_i N_i^{(m)}(\xi) X_i \tag{22}$$

and improves these estimates to approximately

$$\|x_h - x\|_{L^\infty(\Gamma)} = O(\Delta^{m+1}), \qquad \|n_h - n\|_{L^\infty(\Gamma)} = O(\Delta^m). \tag{23}$$

This is particularly relevant for pressure because the normal stress contains the pressure term

$$\sigma(\mathbf{u}, p)n = -pn + 2\eta\varepsilon(\mathbf{u})n. \tag{24}$$

Replacing n by a discrete normal n_h perturbs the pressure contribution by approximately $-p(n_h - n)$. Higher-order curved geometry reduces this normal and boundary-location error, and should therefore reduce boundary pressure errors most strongly where $|h'(r)|$ and $|h(r)|$ are largest. Table 9 summarises the main terms controlling the inner–outer error difference.

Table 9. Diagnostic quantities controlling the difference between inner- and outer-boundary pressure errors. Values use $R_1 = 1$, $R_2 = 2$, and $C = -1$. The listed effects are qualitative because the final error also depends on the discrete pressure field and solver tolerance.

Mechanism	Diagnostic quantity	Benchmark value	Expected effect
Signed pressure mean	$\int_{\Gamma_r} p^* \, ds$	Zero on both boundaries	Does not determine L_2 error because signs cancel only before squaring
Trace magnitude	$\lvert h(R_1)\rvert$, $\lvert h(R_2)\rvert$	$2.328, 0.918$	Larger inner pressure amplitude increases potential inner trace error
Boundary measure	$\sqrt{R_2/R_1}$ in an L_2 norm	1.414	Would favour larger outer error if pointwise errors were equal
Radial geometry sensitivity	$\lvert h'(R_1)\rvert$, $\lvert h'(R_2)\rvert$	$13.31, 0.164$	Strongly amplifies inner error from radial boundary displacement
Tangential phase sensitivity	$\lvert h(R_1)\rvert / \lvert h(R_2)\rvert$ in the $k^2 h \, \delta\theta$ term	2.54	Inner trace is more sensitive to tangential/phase perturbations; sensitivity grows with k
Boundary geometry order	Linear: $O(\Delta^2)$ position, $O(\Delta)$ normal; curved degree m: $O(\Delta^{m+1})$, $O(\Delta^m)$	Method-dependent	Curved elements reduce pressure-trace and stress-normal errors, especially on the inner boundary

First, higher-order mixed discretisations provide substantially improved pressure accuracy relative to lower-order formulations. This is particularly important in geodynamics applications where pressure gradients strongly influence buoyancy-driven flow and dynamic topography.

Second, curved geometries remain sensitive to geometric approximation order. Although linear geometry representation was sufficient to recover theoretical convergence trends for the present benchmark, higher-order curved geometry representations may become increasingly important for extremely fine meshes or strongly curved spherical-shell problems.

Third, equal-order discretisations continue to exhibit weaker pressure robustness relative to stable Taylor–Hood formulations. This behaviour aligns with expectations from the Ladyzhenskaya–Babuska–Brezzi stability condition.

The benchmark therefore provides a valuable verification framework not only for Underworld3, but also for future development of curved-element geometries, advanced boundary-condition implementations, and spherical-shell finite-element formulations.

7 Conclusions

We have presented a verification study of the Thieulot–Puckett annulus Stokes benchmark using Underworld3. The benchmark provides a stringent test of incompressible Stokes flow in curved geometries through the combination of radial gravity, coordinate-dependent forcing, tangential boundary velocity conditions, and non-trivial pressure variation on curved annulus boundaries.

The principal findings are:

1. Higher-order Taylor–Hood discretisations reproduce the expected smooth-solution convergence hierarchy.
2. The $P_2 \times P_1$ discretisation achieves approximately third-order velocity and second-order pressure convergence.
3. The $P_2 \times P_0$ discretisation achieves approximately second-order velocity and first-order pressure convergence.
4. The $P_3 \times P_2$ discretisation achieves approximately fourth-order velocity and third-order pressure convergence before fine-resolution error saturation.
5. Lower-order discretisations exhibit weaker pressure robustness and reduced convergence quality.
6. Underworld3 accurately reproduces analytical Stokes solutions in curved annular geometries.

These results establish the Thieulot–Puckett annulus benchmark as an effective verification framework for Underworld3 annulus and spherical-shell geodynamics applications.

References

Bernardi, C.: Optimal Finite-Element Interpolation on Curved Domains, SIAM Journal on Numerical Analysis, 26, 1212–1240, https://doi.org/10.1137/0726068, 1989.

Boffi, D., Brezzi, F., and Fortin, M.: Mixed Finite Element Methods and Applications, Springer Series in Computational Mathematics, Springer Berlin Heidelberg, ISBN 978-3-642-36519-5, https://doi.org/10.1007/978-3-642-36519-5, 2013.

Brezzi, F. and Douglas, J.: Stabilized mixed methods for the Stokes problem, Numerische Mathematik, 53, 225–235, https://doi.org/10.1007/bf01395886, 1988.

Ciarlet, P. and Raviart, P.-A.: Interpolation theory over curved elements, with applications to finite element methods, Computer Methods in Applied Mechanics and Engineering, 1, 217–249, https://doi.org/10.1016/0045-7825(72)90006-0, 1972.

Davies, D. R., Kramer, S. C., Ghelichkhan, S., and Gibson, A.: Towards automatic finite-element methods for geodynamics via Firedrake, Geoscientific Model Development, 15, 5127–5166, https://doi.org/10.5194/gmd-15-5127-2022, 2022.

Gassmöller, R., Dannberg, J., Bangerth, W., Puckett, E. G., and Thieulot, C.: Benchmarking the accuracy of higher-order particle methods in geodynamic models of transient flow, Geoscientific Model Development, 17, 4115–4134, https://doi.org/10.5194/gmd-17-4115-2024, 2024.

Girault, V. and Raviart, P.-A.: Finite Element Methods for Navier–Stokes Equations, Springer Series in Computational Mathematics, Springer Berlin Heidelberg, ISBN 978-3-642-61623-5, https://doi.org/10.1007/978-3-642-61623-5, 1986.

Guzmán, J. and Neilan, M.: Inf-Sup Stable Finite Elements on Barycentric Refinements Producing Divergence–Free Approximations in Arbitrary Dimensions, SIAM Journal on Numerical Analysis, 56, 2826–2844, https://doi.org/10.1137/17m1153467, 2018.

Hughes, T. J., Franca, L. P., and Balestra, M.: A new finite element formulation for computational fluid dynamics: V. Circumventing the babuška-brezzi condition: a stable Petrov-Galerkin formulation of the stokes problem accommodating equal-order interpolations, Computer Methods in Applied Mechanics and Engineering, 59, 85–99, https://doi.org/10.1016/0045-7825(86)90025-3, 1986.

Kramer, S. C., Davies, D. R., and Wilson, C. R.: Analytical solutions for mantle flow in cylindrical and spherical shells, Geoscientific Model Development, 14, 1899–1919, https://doi.org/10.5194/gmd-14-1899-2021, 2021.

Lenoir, M.: Optimal Isoparametric Finite Elements and Error Estimates for Domains Involving Curved Boundaries, SIAM Journal on Numerical Analysis, 23, 562–580, https://doi.org/10.1137/0723036, 1986.

Scott, L. R. and Vogelius, M.: Norm estimates for a maximal right inverse of the divergence operator in spaces of piecewise polynomials, ESAIM: Mathematical Modelling and Numerical Analysis, 19, 111–143, https://doi.org/10.1051/m2an/1985190101111, 1985.

Thieulot, C. and Bangerth, W.: On the choice of finite element for applications in geodynamics – Part 2: A comparison of simplex and hypercube elements, Solid Earth, 16, 457–476, https://doi.org/10.5194/se-16-457-2025, 2025.

Thieulot, C. and Puckett, E. G.: Incompressible Stokes flow in an annulus: An analytical solution and numerical benchmark, https://www.math.ucdavis.edu/~egp/PUBLICATIONS/JOURNAL_ARTICLES/SUBMITTED/CAPT-EGP-2018.pdf, preprint submitted to Computers & Geosciences, 2018.